PD

(BOM) (PCB) € • , ƒ „
…† ‡ , ˆ ‡

EFFICIENT ALONSO PISTOL DETECTOR (PD) - REAL CLONE EDITION REV 2.1

PD € • ,ƒ ·„

‰ Š ƒ „ ‹ • Pistol Detector ‹• Ž) PD • , • ‘† † ’ • “ ”•‡
–– –– ~ ™ (GD-348 , “ ~ ‡‹ Š• ‹š ‘ † ‘† •
› • ,œ ‘† … ” , € • ž . ‡ …†€ Ÿ € Ž ¡ , ¢• ‡
™ ‘ ™ •‘• ‘ , € • Ž £ˆ • ‹, † ¤£„ –‹• • –‹ ‹•
¥ •†„ ‡ •‡ •™ … ¦ ˆ ¥ ~ ‘ • ˆ ‹,
Induction) § • ¨ “ ‘ ©/“ •‡ € • ª ~ PD ¢• ‡ ‘ ‘
Balance VLF) ” •†, • («) – ¬ ‘ , € • € † Š- † ‡
(Directional Passive LF Ferrite) , “ ª•‡ .
‹ ~ Š- € –•‘• š‘ ~ € : , • ® † ‹ ¯ † ’ ‹• ¢• ‡
Geiger) ”© “ ‘ ‹ † ’ ¬ ž • ° • ± Ž ©
Clicks) • ² Ÿ€ ¤†• ¬ ‘ , € Ž •‡ ‹• … Ÿ‘ • “ , ‡ .
•• ¨• ³ Ÿ ¬ ‘ ‘ ~ ” •†, • ± ‹• … Ÿ‘ • ´ ~ €
• Ž Ÿ, • •‘ ‡ ‹ ¨‘ ¬ ‘ , € ~ € µ • “ Ÿ € • €
. ~ ‡ ¶ š ¡ ° • •

:… † • ‡, ˆ ‰ Š ‹ Œ • Ž

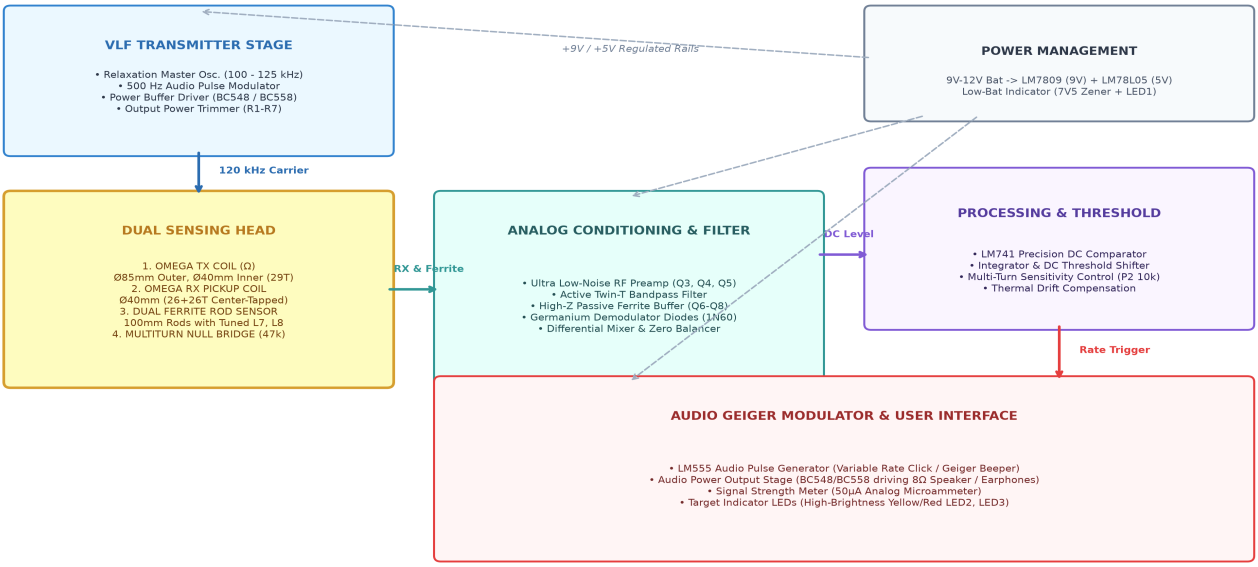
… † • Š ‹ Œ • ˆ ‘	•• / , ˆ ‰
(PD (Alonso Pistol Detector Real Clone Rev 2.1 • , † ’ • ‘† • €	¢• ‡ ´ ‘
‘ ‘ ~ † ~ † € ‹ ª - + (TX/RX) ” ¨ ¬ ‘° • •†,	† ‡ •ˆ ‡
Ÿ ·¸¸ † ’ ¬ …† ‡ ° “ ‹• “ ¹ (VLF) Ÿ † , º»· º¸¸	• ¯ ~ ‡… ‡†‘ , ¬ ‘ , €
¼¸mm ‹• ½ • ‡ ·» “ ‘ © ¾·mm ‹• ¿ · »À “ •‡ € « •†,	” “ ‘ © “ •‡ € •†,
L8 L7 ‘ Ž £, ° • Á ¥†ª‘ • • º¸Âº¸¸ ~ † € ‹	† Š- € † ‡
”‘ ¡ ‘ LED ® ·¸ÃA + Ä Å† ‘• • + …† /†© • ‹• ”© ¬ Æ • µ	€ † ’ ¡ ¢• ‡
(• - LM78L05 (· (LM7809 (À • ¨• † °†© • DC º» À	“ ”•‡ ‹ Á µ • Ç •
(• †É • • ® ‡ ·¸ ~ ² • , Ž È ¢, •) È • ¼¸ »·	€ È … -
…† Ÿ € ¢ž ¯ ‹• ‹• • • · ¼ Ë , ® • º¸¸¸ ~ ² • ‘ ~ ±	š Ë , ® ±

Ž †• ’ “^ ‚‰ ” , € • •^,• – -.˜

† ‡ ” ¨ ¬ ‘° • •†, “ •‡ € ¨„ • “† ‘ ˜ ‡ Š ´ © ¦† • LM555 †’ †° LM741 Ì ª ©‹ ¨ Á©… †•, • € † € : ² ‘ ‹Áµ Í † ^ § ” ‡ ™ª •

EFFICIENT PD DETECTOR (PISTOL DETECTOR) - SYSTEM ARCHITECTURE

Dual Induction Balance Omega Loop + Directional Passive Ferrite Front-End



:… † ‡ – – ’ “^ ‚‰ Ž †• ‚™ ‚• š• ›

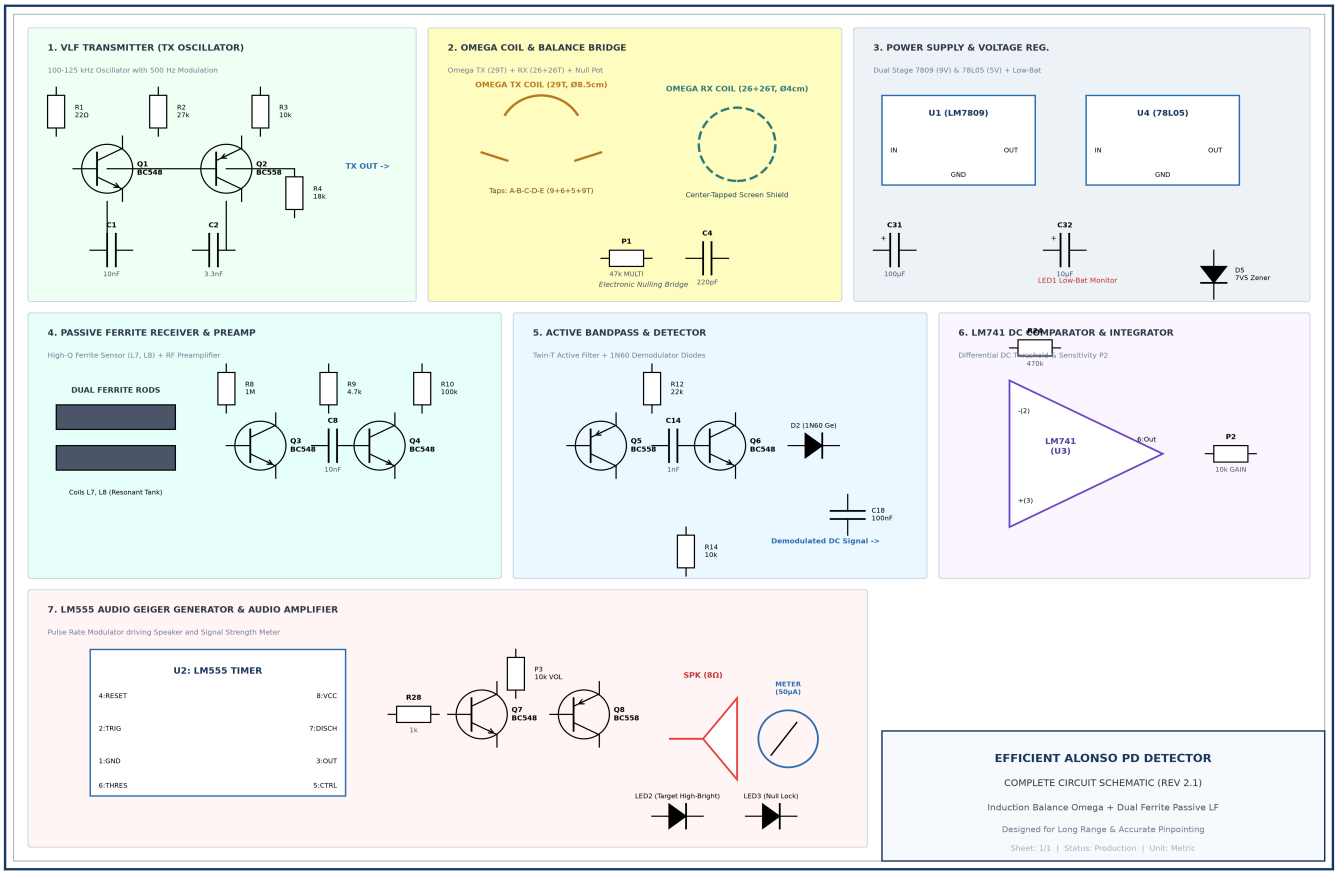
‹ †’ ‹• Ÿ † ‚º» ¸ • ¯ Î† VLF “ •‡ € :(TX) • ¯ Î† § ‡ † Ï ‘ Î† –¢ . ‚ ¢ Ÿ ” •¢†, ‹• “ ‚ † Ÿ ·¸¸ †’ ¬ •“ ÐÑ …†€ ‘ ‡ ¢ - Ò ‡† ‹, ‚ ž¢ …† Ò ¢ . ©

´† ¢ Ó • Ž • É :(Nulling Bridge) ©• Ì ª ••¨• €¢ Ï . ‚ • ª’ ‹• “ ¹ ^ • € Ò ‡† ¨• ¢¨ § ” ‡ P1 … •† . † RX •¢†, Ì ª § ” ‡ † § … †^ ¢ • ¿ -† … ‹• Ž

“ ‚ ¢†¨ ² Ò ‡† •€¢ § ” ‡ : ”¢© ¬ …† ‡ ° …† ‡ • € ¢†¨ Ï †¢ Ò ‡† ˜ ‡ ™ • ˜ ¬ “ “ †® Twin-T Á©… • € ˜ ¢†¨ Ÿ¢†‘¢, · ‹¢ Ç • ½ Ô• - ^ . † LM741 ©‹ ¨ ‹• DC Ç • 1N60 ´†‘ Ç ‹• ¨® • Ž ‘ †© • ‹• †’ ¬ Õ ‘ “ µ ··· ¢ . ² ¢ Ÿ€

PD • › š œ• • ž • ‘ .Ÿ

… ¡ ‘ PD † ‘† • • 𝔼 “ ‡ Š ˜ • ‘ •‡ ‹š ‘ 𝔼˜ ‹ ¡ ¨‘ ‹™ Ÿ𝔼†‘¢, †• 𝔼Ÿ‘ – 𝔼• ‹𝔼 Á©“ . “ ¶ š ¡ • Æ 𝔼 †• ‘ Ö x† ‡ § ‘ ¬ ‘° • • ‘ …˜ ^ Øº ¨ : ‡



:œ• • † ^ š … , ^ • Š

(Q2 (BC558 PNP (Q1 (BC548 NPN • ™ †• 𝔼Ÿ‘ ‹𝔼 • : 𝔼 VLF ˜ ‡… ‡†‘ Ï
(C2 (3.3nF (C1 (10nF …˜ ^ . , • ® 𝔼 – ¡ ™𝔼 ˜ ‡… ‡†‘ † ’ ‹•
¬ ‘ Ė (R1 (22« ¨ ‘˜ ‡ 𝔼 Ÿ † , º»¸ ¯ ‹𝔼 ¬ ‘ , €
. , ¢ Ó •𝔼†, ‹• 𝔼 Ÿ
(º¸ ¢ P1 (47k … • † • † ‘ • :(Electronic Nulling) ™ ‘ •™ § ‘ • Ï
•𝔼†, – • ³𝔼 Ÿ Ç • –𝔼 †, ‘ †•• † ‡ “ ‹ ¢ ´ ’ •
. ‘ ‡ • (Dead Hole / †™‡ “ ª¯) ª ’ ‹ ¨‘ ‹• ¨ “ ‘ © “ •‡ €

• , ¡^ ^ Š ‘ ¢ ‹ ‹Œ • ‹ £ ¤ , • .¥

:(Passive Ferrite Stage) Ÿ€†‘¢, “ , €†¨ ² † € “ ‘ © ‹¨ „ .º
€€ –•‘• … † ® ‹• (Ãi Ù 600) ° • ‹ „ µ ¥†ª‘ ¿€ Ì • ˜† € ‹
‹• L8 L7 Ö ¢ ‡ . , • ® …†€ Ÿ € ˜ ‘ “Ÿ ‘†€ … Î †
…† “ •‡ € ¬ ‘ , € • ¨ (1nF + 1.8nF) ¬ ™ € €•‡ ˜† …˜ ^ “
º ¨) ° • • ¬ ‘ Ê • Q4 Q3 Ÿ€†‘¢, †• €Ÿ‘ . ‘ “
‹¨ „ ‹• “ , €†¨ Ÿ€†‘ ‘ • ¯ • • ™ Ì § ” ‡ (R8 ¢ ”
. € ‘ § ‡ • €

:(Active Filter & Detector) ´† ‘ Ç †€ ˜ ‡ ™ • †•, Á©… • € .»
§ € • € C14 …˜ ^ R12 R14 • ¨ ‹™ “ ‹• Q6 Q5 †• €Ÿ‘
¤ • ¬ ‘ , € ¢¯Ÿ ‘† ‹• • € –€ . • ™¡ Twin-T Á©…
˜ ‡ ™ • †€ ¬Ê‡ . †® • ¯ ¬ ‘ , € Ò¨€ “ , ŽÁ¯ (Ÿ ·¸) Š
‹• § ” ‡ Ë † (¸.» – € ‹‘•‡• Ç • • AA119 €1N60 §) D2 ´† ‘ Ç
‹ ªÈ • , †„ ‹• RF •ÊE (C18 (100nF …˜ ^ . , •€ DC µ• Ç •
. € ‘

DC Ç • :(LM741 (DC Comparator ½ Ô• • DC Ú ‡ ©§ ”•‘ Ì ª ©‹ €¨ .Ä
. © § ® LM741 Ì ª ½ Ô• (» ‹€) “ ‚‰†™ ‹• “ Ž ’
• È• · ¬ ‘ € Ç • ‹• (P2 (10k ‡ ¯ • † ‘• Ò ‡ † (Ä ‹€) “ ‚‰†™ É
. , – DC – © ¯ € (R24 (470k ª • € ¨ . ‡
- ^ € Ç • ‹• Ž ¬ ‡ ˜ ‘ • ‘ • Ž£•^ ™ • ½ Ô•
. ˜ ‡ •€ Û ‹€

:(Audio Geiger Generator) - ^ †€ LM555 € • ”€© †’ ¬ † .¼
½ Ô• - ^ Ç • . , • ® •••‡• ˜ ‡… ‡†‘ (U2 (LM555 € €•
¬ ‘° • ¯ . ‡ • È• ··· €• (· ‹€) Ç • § • , ‹€ ‹• ¨•
· ‹€ Ç • Ž ¬ ‡ •. , † (‹ ‘Þ º ¯) ´ • †’ ¬
¤†• ‹• •€ ² €Ÿ€ ”€© “ ‘ ‹ •• • ¬ ¬ ‘ , € “ , µ
‹€Áµ Ê • ™ ·¸ • ¢ ¾ †© • Q8 Q7 †• €Ÿ‘ . † •€ •
. ,

• :(Power Management) • †• ‘ ”‘ ¡ ‘ ‹‘© •¨• Ç • ¹ ¯ .·
: ‡ “ “ ª•‡ •¨• † °†© ˜ ‡ ¯ € ¬ ‘ , € ² ‘ ˜ ©† -
, – Ñ ½ Ô• • ‘ € À Ç • (U1 (LM7809 ’ † °†©
©‹ €¨ •- … † ® ‹• · Þ • “ ¤†€ Ç • (U4 (LM78L05 † °†©
Ç • € • Q11 Æ Ü ›† ‡ †• €Ÿ‘ (D5 (7V5 ‘˜ †€ . ˜ ‡ Š Æ ‡ €•
. , – (LED1) • ¡ § ¾.» €˜ ‹• •

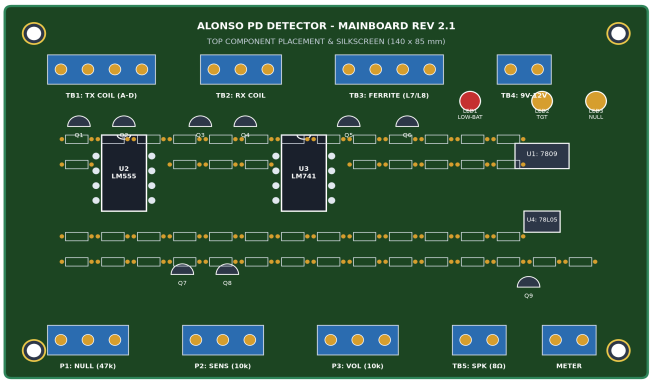
(Bill of Materials - BOM) • , ¡^ Š ƒ ¦ § š •¡ .¨

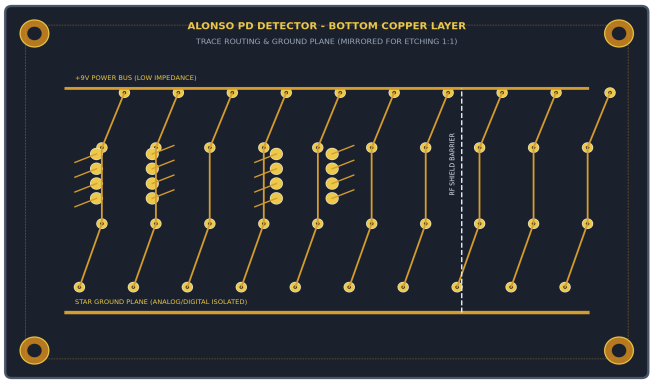
• © • , È š ¡ ³ ™ ¨ ‹ ‡ ,¥ • ℡ ˜ § - ‹ ,
Ÿ℡ ° • ...˜ ^ ¨ • Øº ¬ ‘ • • †É ~ “ ª• ‡ . ‘ “
MKT/MKP ‡ Ÿ ‘ ...˜ ^ •:

©	ž •	žƒ ¦ § ^ ‘ •	ª / «• ‰	^ ƒ	^ ,- Š ‹ Œ •
1	U1	LM7809	TO-220	1	(1A) ¹ “ ¹ À Ç • ^ † °†©
2	U2	LM555 / NE555	DIP-8	1	DIP ,† ‡ • ”℡© † ’ ¬ † ° ℡
3	U3	LM741CN	DIP-8	1	• †É ,† ‡ • Ÿ℡†‘¢, Ì ª ©‹ ℡ ¨ ½ Ô•
4	U4	LM78L05	TO-92	1	(100mA) ¹ · ℡ ¤†€ •- Ç • † °†©
5	Q1,Q3,Q4,Q6, Q7,Q9-Q11	BC548 (Ýþ 2N3904)	TO-92	8	(Hfe > 200) ° • ¬ ‘ , € Ÿ℡†‘¢, NPN †• ℡Ÿ‘
6	Q2,Q5,Q8	BC558 (Ýþ 2N3906)	TO-92	3	’ - ^ “ , ℡† † ® PNP †• ℡Ÿ‘
7	D1, D3, D4, D8	1N4148	DO-35	4	(High-speed) ‘†™ ‡ •℡ ‡ Æ Ü x†‡ †℡
8	D2	1N60 / AA119	DO-7	1	¸.»V Ç • € • ´† ‘ Ç RF ˜ ‡ ™ • †℡
9	D5	Zener 7.5V 0.5W	DO-35	1	• Ç • € ¡ ¬ ‘ € ‘˜ †℡
10	LED1, LED2, LED3	High-Bright 3mm	Radial	3	(§ ‘ ¬ ‘° • :LED3 Ž :LED2 • :LED1) †‘ §
11	P1	47k« / 50k«	Multiturn 10T	1	ª ’ ‹ ¨‘ ¬ ‘° • Š- Ÿ‘ †• º ¸ ... • † • † ‘ •
12	P2, P3	10k« Linear	Panel Pot	2	(’ • :P3 – © ‡ ¯ :P2) • § • , ´†
13	R1	22 «	0.25W Metal	1	• ‘ ˜ ‡ ... ‡ †‘ - ^ ¬ ‘ Ê ¨
14	R2, R4, R6, R7	27k, 18k, 3.9k, 8.2k	0.25W 1%	4	“ •‡ € † £ ‡ • ¨ ¢ ¨ ‰ ℡ • ‹™
15	R3, R5, R14, R16	10 k«	0.25W 1%	4	℡ Ç • ¢ ¨ †•™ , • ¨
16	R8, R43	1 M«	0.25W 1%	2	℡ € † ‡ ß ´ ® Š- ” ¨
17	R9, R11, R28	4.7k, 1k, 2.2k	0.25W 1%	3	§ ... ℡ - ¬ • Ç £ †, ¨
18	R12, R19, R24	22k, 100k, 470k	0.25W 1%	3	LM741 ½ Ô• • € †•, • € ¨
19	C4	100pF - 470pF	Mica / C0G	1	(Silver Mica) ª ’ ℡ ¿ ℡ Ì • § ‘ Ç £ †, ℡ ...˜ ^
20	C11, C22, C32	1 nF (1000pF)	MKT 63V	3	• € ℡ € ‘ ¬ ‘ ‘ ˜ ¢ € Ÿ℡ ° • ...˜ ^
21	C2, C23	3.3 nF / 1.8 nF	MKT 63V	2	Ç £ †, • ¯ ¬ ‘ , € ¬ ‘ ‘ ˜ Æ ‘† ...˜ ^
22	C1, C8, C24, C27, C30	10 nF	MKT 63V	5	RF ‰ • • € ¨ „ Ç £ †, ...˜ ^
23	C18, C29, C38	100 nF (0.1ÃF)	Monolithic	3	†•™ € ’ ℡• ‹℡Áµ Æ †, ...˜ ^
24	C31, C32, C34	10ÃF, 47ÃF, 100ÃF	Electrolytic	3	(, »·V • ¯) ‹℡Áµ € ’ • •™ ...˜ ^
25	Omega TX	Wire 0.30mm (29T)	Custom Coil	1	- ^ ¿ · • ” Ž £, • ¸.Ä, “ ‘ ¬ ,° ¢ ‡
26	Omega RX	Wire 0.20mm (52T)	Custom Coil	1	• ℡†€ • (»Û+»Û) ½ • ‡ ¸.»¸ “ ‘ ¬ ,° ¢ ‡
27	Ferrite Rods	100mm Â 10mm	High-Mu Rods	2	° • „ µ ¥†ª‘ ¿ ℡ Ì • • †É ℡†℡ ℡ € ‹
28	SPK / Meter	8« 0.5W / 50ÃA	Panel Meter	2	Ê • ™ · ¸ ‹• ¨® • “ ‹• ¢ ¾ † †© •

,– ¬ ‡^ (PCB Layout) ‰ - ^ , • Ž • ‘ .®

‘ © • “ •‡ † ’ ‹• • ¾· º¼¸ ‘•‡ • ℉˜ ƒ „
Top Silkscreen) … ‹℉° ½ ‡ ℉†È . ‡ “ ¯ „ “
Overlay) • ℉˜ ‡ ‡ ℉†È (Bottom Copper Layer) † ’ ‹•
: ‹ˣ , ‡ † ¢ ¨• Ô ¿ ‡ (Mirrored) “ ‰†™ º:º ‰ ¨





:, • ¬ ¯ , ° ‡ ^ ^ ‡š ƒ¡^

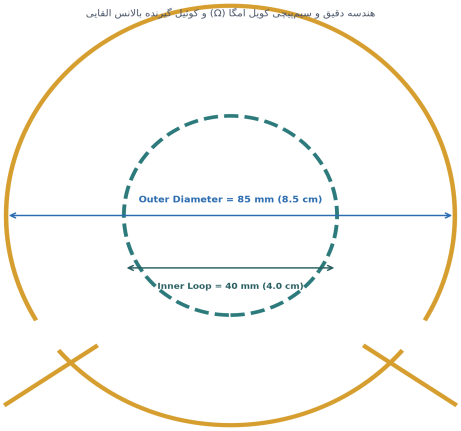
Ä· ¬ š Ì • FR4 ‰£© ℉€ €˜ † , ‹ ’† : € ¬ - .º
˜• Ò † •†„ • Á- • ‹• † ‘ †š•‡ € . † “ ª•‡ … ™
. ‘† § ‘ ¬ ‘° • § … †^ ¢ ‹• ¨℉® ¨ µ à® • … • •
†© • … ℉ - ¡ ©˜ • :(Star Grounding) ‘ © § ” ‡ ‘ © ™ª .»
ª ‹ ¨‘ Š “ Ÿž £ , ‰ ¯ “ , ℉†¨ ² ‘ ©˜ . . . ℉
. † ¡ ‘ ¨ ‰ ¯ ‹• Æ Ü ˣ† ‡ Ÿ℉†‘ ‘† • È• ”℉ ™℉ ‹• •
‹¨ „ ™ª ˜ “ … ¡ ‘ PCB ‹ ¡ ¨‘ – ‹ ¨‘ Ò^ :‹Óª …† ‡ ° Ÿ℉ .Ä
–℉ – • †, † † • “ †℉ ℉ † ‹ ’† . ‡ †’ ² š• ˜ ° • ¬ ‘ , €
. † ¢ • • ‹¨ „
• ™ • ℉ • € –℉ ℉˜ Ú ‡ ,¢ … ℉ ˜ ¬ :Ç •‘† ˜ ¬ • † ¡ • .¼
•℉˜ ¨ ‹ , ¢ –É “ ‘ • † ‹• £ , ØÀÀ • Ÿ℉
. © ˜ , £ , , ž℉ ‹•‡ †^ ‘

, † ^ Š ±•‰€• ¯ • •Ž–• • ‡^ .²

• ® . • PD • “ Ė ¿ € Š- † ‡ («) ” •†, “ ,– ß ” € ™ ‘ ™ ‹ ’ € •‡ •™ Æ ‡ : ‡ ™ ® ° • •

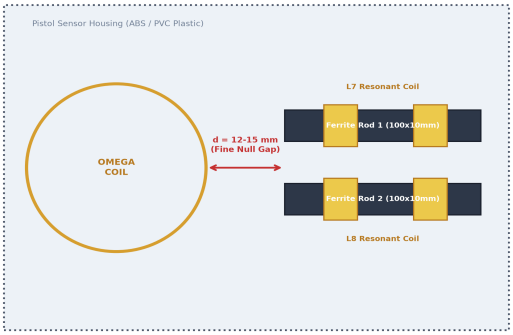
OMEGA TX & RX COIL GEOMETRY

هندسه دقیق و ابعاد سیم‌پیچی کویل امگا (Ω) و کویل گیرنده بالای الفایی



DUAL FERRITE ROD & HOUSING ALIGNMENT

چیدمان سنسور فریت جهت‌دار: زاویه محور صفر و فاصله نسبت به کویل امگا



TX COIL SPECIFICATIONS:

- Wire Gauge: 0.30 mm (AWG 28-29) enameled copper
- Total Turns: 29 Turns with 5 taps: A (0T) - B (9T) - C (15T) - D (20T) - E (29T)
- RX Pickup Coil: 26 + 26 Turns (52T total) of 0.20 mm wire, Center-Tapped
- Shielding: Graphite spray or Aluminum foil (leave 5mm gap to prevent shorted turn)

FERRITE SENSOR & CALIBRATION TIPS:

- Ferrite Material: High initial permeability (μi >= 400 - 800) MnZn / NiZn rods
- Tuning: Coils L7 & L8 wound with 0.15mm wire (80-120 turns) tuned with 1nF+1.8nF caps
- Mechanical Adjustment: Ferrites mounted on brass/nylon micrometer screw for distance nulling
- Crucial Rule: Keep batteries, speaker, and aluminum handle > 25 cm away from coils

:(Omega Coil) † ^ Š - ±•‰€• Š™^,

‘ ‘†€ Ž ¯ ´ € …† • ‰£© ™ € ¡ ^ • † ‹•Š € • :…† • ©¿ Ï ‹€ • • ¼¸ ‹• ^ ‹¨ ¯ • ¾· - ^ ‹¨ ¯ ‹• («) ” . † “ Ë • ‹- ¼· ‹€ ~ - ^

»À ^ † ™€ † ’ ‹• • ¸.Ä¸ “ ‘ ,° ¢ ‡ • :(TX) “ •‡ € •€†, Ü ¢ ‡ Ï <- (À ˜ ¬) B ‹€ <- (¸) A ‹€ : ¿ Î Š•‡ ¿ . † “ Ü E ‹€ <- (»¸ / ”€ · ˜ ¬) D ‹€ <- (º· / ”€ Û ˜ ¬) C ‹€ .(»À / ^• À ˜ ¬)

“ € Ž£, ” ^ ‹¨ ¯ Ÿ, :(RX) € ¨ ¬ ‘° • “ ‘ © •€†, Ü ¢ ‡ Ï ½ • ‡ •, ·») † “ Ü • ¸.»¸ ,° ¢ ‡ • • ¼¸ ‹• • ™ .(»Û - ^ •

† † • •€†€ †‘ • “ ‘ © Ž£, :(Faraday Shield) € •‡ •™ Ï ˜• • · €® €† Ž ™ € •¯ . † “ ‘ † € © Ė‡ € ¦˜ ‘ ‹• “ ¢ •€†€ š –€ ¢ ‡ . † ¡ ‘ • ™ ¡ “ †, § È ‹¨ ¯ ‘ • . † •È• ‘ ©

», ¨€ ‹ ’ € • • º¸Âº¸¸ ˜ † € € ‹ :† € € † ‡ Ç •‘† Ï –• ” ‹ € € ‘ ¡ †„ ‹ ’ €. ‘† ‹ ” •€†, ¡ • . © ¢ Ó • ™ Ÿ € É ¢ Ó Ö • ‡ • º. º»

”^ • ^,‰^ ‡^ ,£ ž ¦ ‘ ” • ^, •¡ Š ,‰ .³

:(Nulling Procedure) ,£ ž ¦ ‘ ´•§ µ ¶ – €• · š™^,

. ‡ PD • ℮ Ÿ ”‘ ª” ™ ® ’ ˜ (True Null) ª ’ ‹ ¨‘ ‹• … ‡
“ ”•‡ :(‹ ™ ‘ ™ ¬ ‘° •) º ´ © : † ´ ž‘ “ ‡ ´ © ‹‡ …† ‡ ,
†„ † ℮ € • ™ ¢ Ó Ö •. , – Å Ÿ• Ÿ € … •
AC Ç • µ • ‹• ” •℮†, ‹• ‘ ℮ € ‹
“ ª•‡ •:(™ ‘ •™ ℮ á ¬ ‘° •) » ´ © . ‡ • †^ ¯ –℮ • , ‹• “ , ℮†¨ ²
℮ . ‘ ^ Ü• ´ • • (P1 (47k … • † • † ‘• ℮ ® • †©Ö ˜
†‘¢, ℮ Ë † ^ (LED3) § ‘ •ª § “ , Í †¨‡ ª ’ ‹• • ‹• ¨® â ^ ‹ ¨‘
† ’ ‹‘•‡• ¢ Ó) Ä ´ © . ´ ‘ (Dead Hole) › †™‡ “ ª¯œ ‹ ¨‘ –℮ . †
”℮© ´ • ‹, ‘ ˜ , ˜• (P2 (10k ‡ ¯ ´† …† , :(†
• , © • ‡ ¯ –℮ °• “ ”•‡ …† , . † “ (‹ ‘Þ º ¯)
. Ë , •

:(Field Operation Guide) “ž ¦ ‘ “ª ƒ • ”^ • Ž ^

¨€ “ ”•‡ :(Radial Scan) Ž •‡ – ‹- ÄÛ¸ †ž• - :º ˜ € Ï
¨ “ ”•‡ ¦†‘ ‹, ” . ^ Ü• • , “ ℮ ℮ • ‹• . ℮ ”• ‹ ‡ ± ª
‹•€ © • • ƒ † Ì ‹• ”℮© Æ • © ℮ …†€ Ÿ € ℮ •‡
. , Á© £® ‘¿ • ‹℮ ˜ –℮ . † Ž ‡ ‹• • ‹• ¨®
• ‹• Ì € Ò^ :(Triangulation) •à ¹ § ” ‡ , ¯ :» ˜ € Ï
•. † ‹•‡† ¤†• ¬ ‘ , € Ž ‹• … ℮ Ÿ‘ •. , , ¯ † - ‹•
•„ ¨ â ´ ž‘ •℮•‡ • ·¸ ‹ ’ € • ‹ ¨‘ ˜ • ¯ ^ ˜ ©† -
. ‡ Ž “ Ò^

Ÿ, • » ‹ ’ € :(Pinpointing) ” •℮†, • • •‘ ‡ ‘˜‹ ¨‘ :Ä ˜ € Ï
•℮†, ℮ ¨ … . , ¯ • - † ’ ‹• ´ • ℮ ”• – ˜ ‡ ‹• “ ”•‡
. † •℮ • ℮˜ ¤†• ‹• ’ “ , • ℮ Æ †, …†€ Ÿ € • ”
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